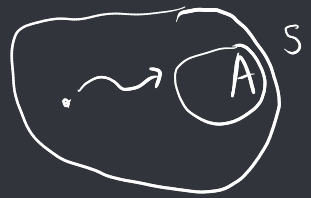
Today: - reversible MCs continued:

- Metropolis algorithm for Hamiltonian systems:
  Glauber dynamics for the Ising model.

- hitting probability & hitting time:

hitting probab: probab. to hit A before B.



expected hitting time of A from different starting points.

Glauber dynamics: given state space S

(for Ising model on big square,

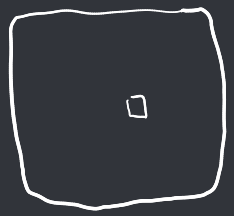
$$S = \{ f : \{1, \dots, N\}^2 \to \{0,1\} \}$$

configurations of vertex set $\{1, \dots, N\}^2$.)

suppose $\exists$ function $H : S \to \mathbb{R}$

(Hamiltonian / energy function)

and an irreducible transition probability q with $q(x,y) = q(y,x)$ $\forall x, y$.



(for Ising model, take q:
choose vertex unif. @ random
& flip its state)

Goal: build MC with stationary distribution

$$\pi(x) = e^{-\beta H(x)} / Z \qquad Z = \sum_x e^{-\beta H(x)}$$

by modifying trans. probab. q.

Trick: if $H(y) \leq H(x)$ let $p(x,y) = q(x,y)$

if $H(y) > H(x)$ then $p(x,y) = q(x,y)\, e^{-\beta(H(y)-H(x))}$

(& leftover probability added to $p(x,x)$)

Think:
- q "proposes" a step
- if step lowers energy, accept w.p. 1.
- " " raises energy, " " $e^{-\beta(H(y)-H(x))}$

Does it have $(*)$ $\pi(x)\,p(x,y) = \pi(y)\,p(y,x) \quad \forall x,y$?

Only need to check $y \neq x$.

If $H(y) = H(x)$

$$(*): \quad \frac{e^{-\beta H(x)}}{Z}\, q(x,y) \overset{?}{=} \frac{e^{-\beta H(y)}}{Z}\, q(y,x) \quad \checkmark$$

$$\underbrace{\qquad\qquad\qquad}_{= \text{ by assump.}} \qquad \underbrace{\qquad\qquad\qquad}_{\text{equal by assumption}}$$

If $H(y) > H(x)$,

$$(*): \quad \frac{e^{-\beta H(x)}}{Z} q(x,y) e^{-\beta(H(y)-H(x))}$$

$$\stackrel{?}{=} \frac{e^{-\beta H(y)}}{Z} q(y,x)$$

on LHS: $e^{-\beta(H(x)+H(y)-H(x))} = e^{-\beta H(y)}$.

so again LHS = RHS.

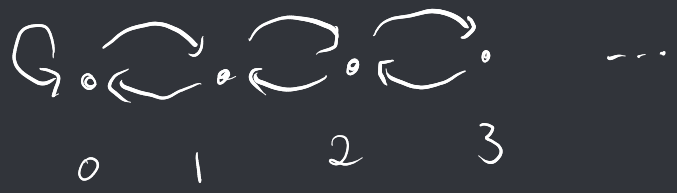
In this framework, $\beta > 0$ is "inverse temperature".

Birth and Death chains

Birth & death MC has $S = \mathbb{N}$

and $p(x, y) = 0 \quad \forall \; |x - y| > 1.$

(no leapfrogging)



ex: X_n population size after n events.

@ each event, $+1$ w.p. p

-1 w.p. $1-p$

$p(x, x+1) = p$ $\qquad$ $p(0,0) = 0$

$p(x, x-1) = 1-p$

Every (irreducible) birth & death chain that has $E_0[T_0^+] < \infty$ has a reversible stat. dist.

How to compute it?

Detailed balance:

$$\pi_n p(n,n+1) = \pi_{n+1} p(n+1,n)$$

$$n \rightleftarrows n+1$$

$$\pi_n = \pi_{n-1} \frac{p(n-1,n)}{p(n,n-1)}$$

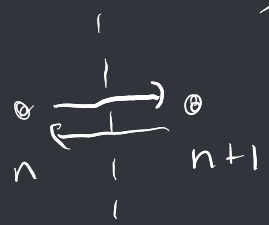
$$\vdots$$

$$= \pi_0 \underbrace{\prod_{i=1}^{n} \frac{p(i-1,i)}{p(i,i-1)}}_{r_n}$$

Does this work?

(exercise: as long as $\sum_n r_n < \infty$, the given π has $\pi = \pi p$)

actual reason:



The only way between $\{0,\dots,n\}$ & $\{n+1,\dots\}$ is to pass from $n \to n+1$ & $n+1 \to n$.

this is detailed balance

$\therefore$ long-term frequency of $n \to n+1$ crossings $\left(i : (X_{i-1}, X_i) = (n, n+1) \right)$

$=$ " " " " $n+1 \to n$ crossings.

since frequency of $(x \to y) = \pi_x p(x,y)$

When does π exist? $\Leftrightarrow$ when is $E_0[T_0^+] < \infty$?

When π exists, $\pi(0) = 1/E_0[T_0^+]$

Above: $\forall n, \pi_n = r_n \pi_0$

$$r_n := \prod_{i=1}^{n} \frac{p(i-1,i)}{p(i,i-1)}$$

Normalization: need

$$\sum_n \pi_n = 1$$

$$\pi_0 (1 + r_1 + r_2 + \cdots) = 1$$

$$\pi_0 = \frac{1}{1 + r_1 + r_2 + \cdots}$$

$$E_0[T_0^+] = 1 + r_1 + r_2 + \cdots$$

ex:

$$p(i, i-1) = 1-p \quad ; \quad p(0,1) = 1$$
$$p(i-1, i) = p$$

$0 < p < 1$.

for which p does $\underline{\pi \text{ exist}}$?

require $\pi_0 > 0$, or

$\Leftrightarrow E_0[T_0^+] < \infty$.

with given values,

$$r_n = \left(\overbrace{p/1-p}^{a}\right)^n$$

$$E_0[T_0^+] = \sum_{n=0}^{\infty} a^n \quad \text{converges if } |a| < 1$$

$$p < 1-p$$

increase prob. < decrease prob.

for case of translation-invariant increment distribution,

stat. dist. exists $\Leftrightarrow$ decrease more likely than increase.

challenge problem: find a finer threshold for existence of $\tilde{\pi}$

by considering

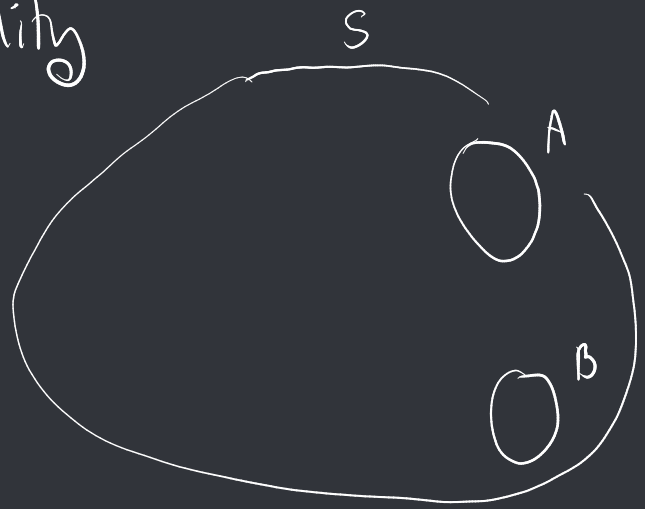
$$p(n, n+1) = \frac{1}{2} - a_n$$

where $a_n \to 0$.

Which (a_n)'s give $E_0[T_0^+] < \infty$.

☆☆

Hitting probability



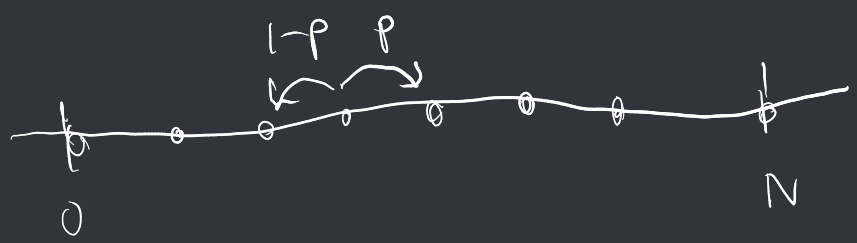
$$T := \inf\{n : X_n \in A \cup B\}$$

Let $h(x) = P(X_T \in A \mid X_0 = x)$

$$= P_x(X_T \in A)$$

probab. hit A before B, starting from x $\quad (A \cap B = \phi)$.

ex: Gambler's ruin.



$$S = \{0, \dots, N\}$$

$$p(0,0) = p(N,N) = 1$$

$$p(n, n+1) = p \qquad 0 < n < N$$

$$p(n, n-1) = 1-p \qquad 0 < n < N$$

$A = \{N\}$ "win" state

$B = \{0\}$ "lose" state

$X_T \in A$ reached goal of $\mathbb{N}$

$X_T \in B$ go broke.

The function $h(x)$ satisfies boundary-value problem

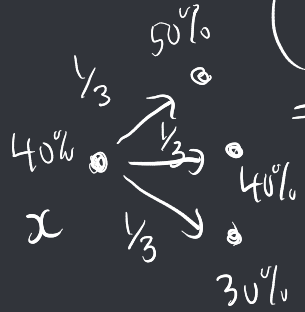
$$h(x) = \begin{cases} 1 & x \in A \\ 0 & x \in B \end{cases}$$

boundary values

$x \notin A \cup B$,

system of linear equations

$$h(x) = \sum_y p(x,y)\, h(y)$$



(probab. to hit A) = weighted average after one step.

for Gambler's ruin:

$$h(0)=0$$

$$h(N)=1$$

for $0<x<N$, $h(x) = p\,h(x+1) + (1-p)\,h(x-1)$ (*)

for $p=\frac{1}{2}$, $h(x) = \frac{1}{2}\left(h(x+1)+h(x-1)\right)$

$\therefore$ h straight line ; w/ BCs, $h(x) = x/N$.

for $p > 1/2$: for any p re-write $(*)$ as

$$p\underbrace{\left(h(x+1) - h(x)\right)}_{\Delta h(x)} = (1-p)\left(h(x) - h(x-1)\right)$$

$$\Delta h(x) = \frac{1-p}{p}\,\Delta h(x-1).$$

suggests solution

$$h(x) = \left(\frac{1-p}{p}\right)^x \quad \text{(not quite).}$$

$\vdots$ wait until later